

OIL India — Production & Exploration Dashboard

Mockups & predictive-analytics concepts — SME verification pack

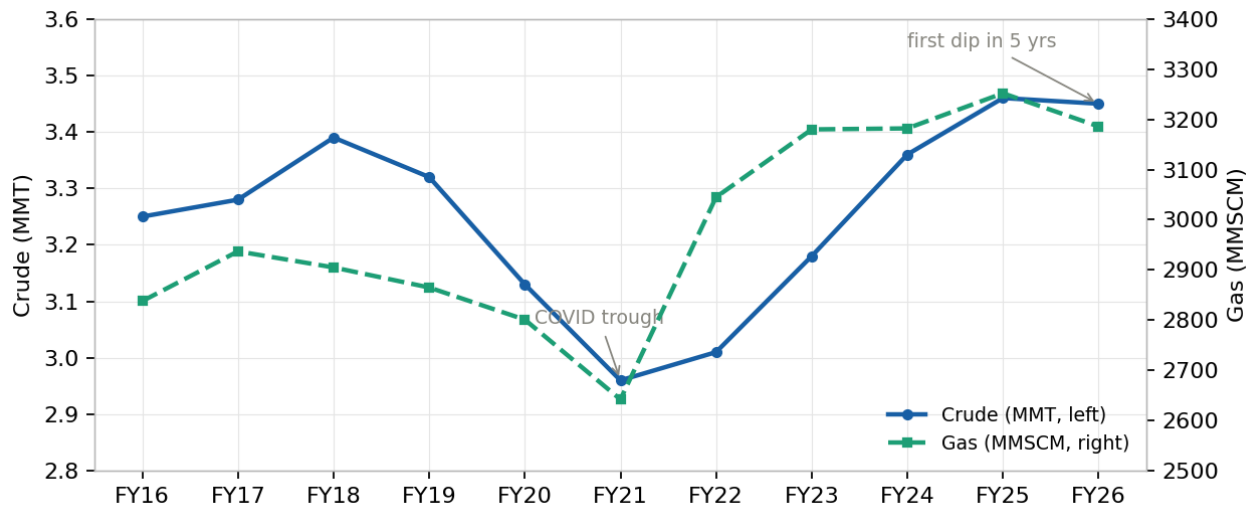
Prepared June 2026 · For internal review · All figures from project files (10-yr Production & Reserves, Workover & Drilling 5-yr, FY25-26 MIS, Annual Reports FY21–FY25) and oil-india.com / exchange disclosures.

Status of forecast content: every chart panel marked "FORECAST" or "illustrative" contains placeholder scenario values for calibration — no model has been fitted yet. Historical panels contain actuals only.

1	Production page — KPIs & 10-yr trend
2	Reserves analysis — RRR & 2P divergence
3	Production forecast fan (FY27–28)
4	RRR scenario fan (FY26–28)
5	Gasification crossover projection
6	Exploration page — KPIs & analytics
7	Exploration effectiveness & required-wells inversion
8	Andaman scenario module (unbooked)
9	Cross-page predictive logic map
10	Predictive claims inventory & SME checklist

1 Production page — KPIs & trend

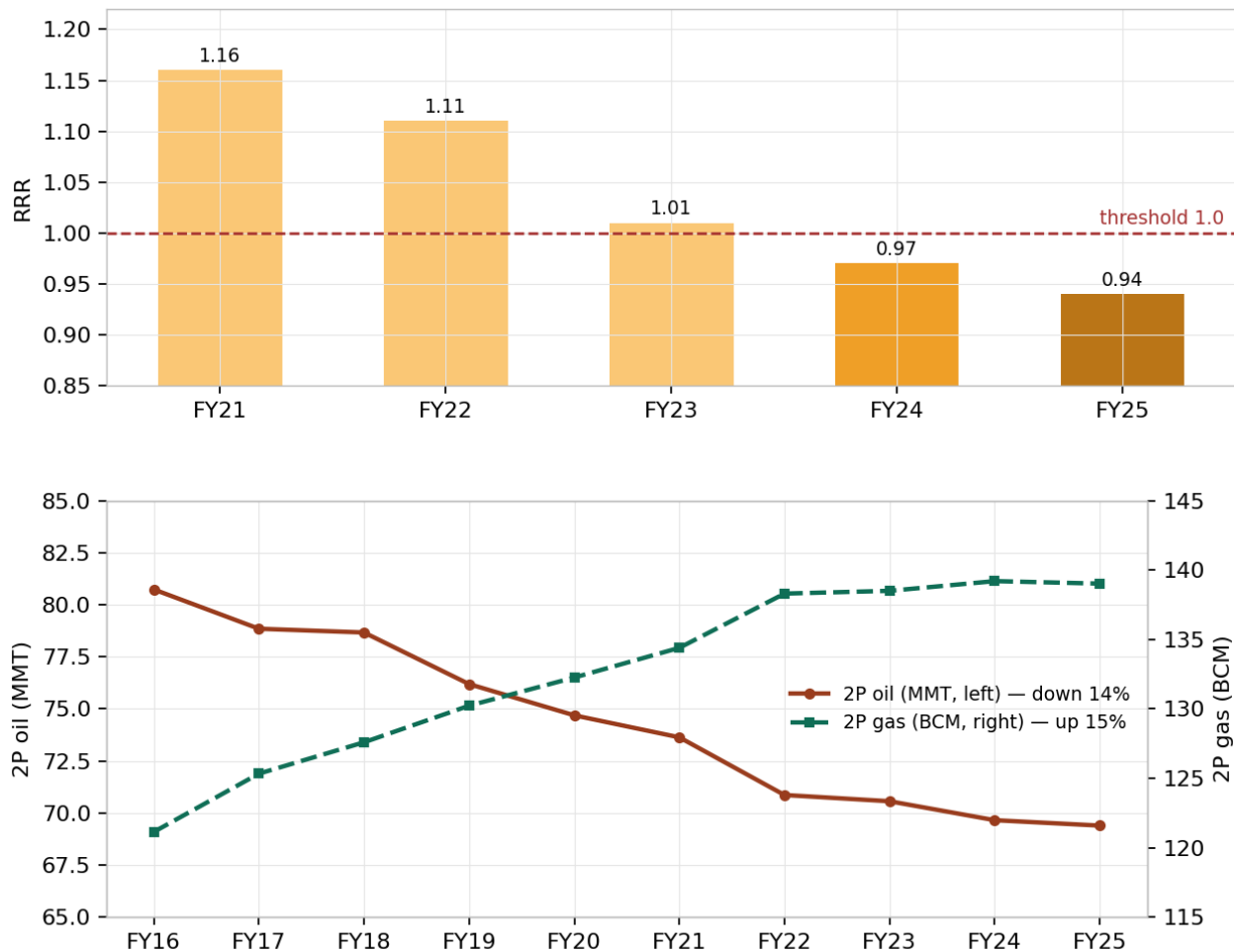
Crude oil production (FY26) 3.45 MMT -0.3% YoY · first decline in 5 yrs	Natural gas production (FY26) 3,186 MMSCM -2.0% YoY	FY26 target achievement 90.9% crude 87.1% gas (with JV) · live MIS	Reserve replacement ratio (FY25) 0.94 below 1.0 for 3rd straight year
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Insight 1 (deep-dive behind this page): recovery from the FY21 COVID trough (+17% to FY25) has plateaued — FY26 printed the first decline in five years. Drill-down: state-wise contribution (Assam base decline vs Rajasthan/JV delivery); FY26 monthly actuals vs BE trajectory (gas at 85.6% of Assam target).

SME checks: ① FY26 dip real, not a reporting artifact? ② Is 90.9%/87.1% the right cumulative achievement to headline from the March-2026 MIS? ③ Crude/gas unit conventions confirmed (MMT, MMSCM)?

2 Reserves analysis — RRR & 2P divergence

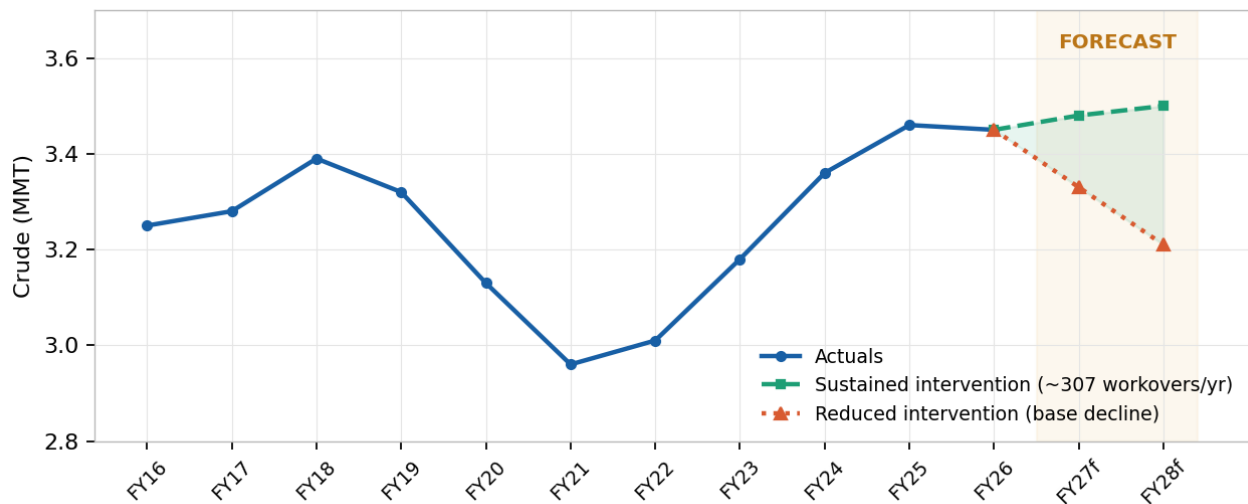


Insight 2: RRR below 1.0 for three straight years; 2P oil down 14% over the decade while 2P gas rose 15% — the portfolio is structurally gasifying. Derived Reserve Life Index: oil ~20 yrs, gas ~43 yrs at current production.

SME checks: ① Does the Excel RRR match the company's official definition (annual vs 3-yr rolling)? ② Are Reserve Life Index figures (~20 / ~43 yrs) directionally right? ③ Does 2P gas flattening in FY24–25 weaken the gasification framing?

3 Production forecast fan — FY27–28

Predictive claim #1 · decline-curve + intervention regression · ILLUSTRATIVE, model not yet fitted

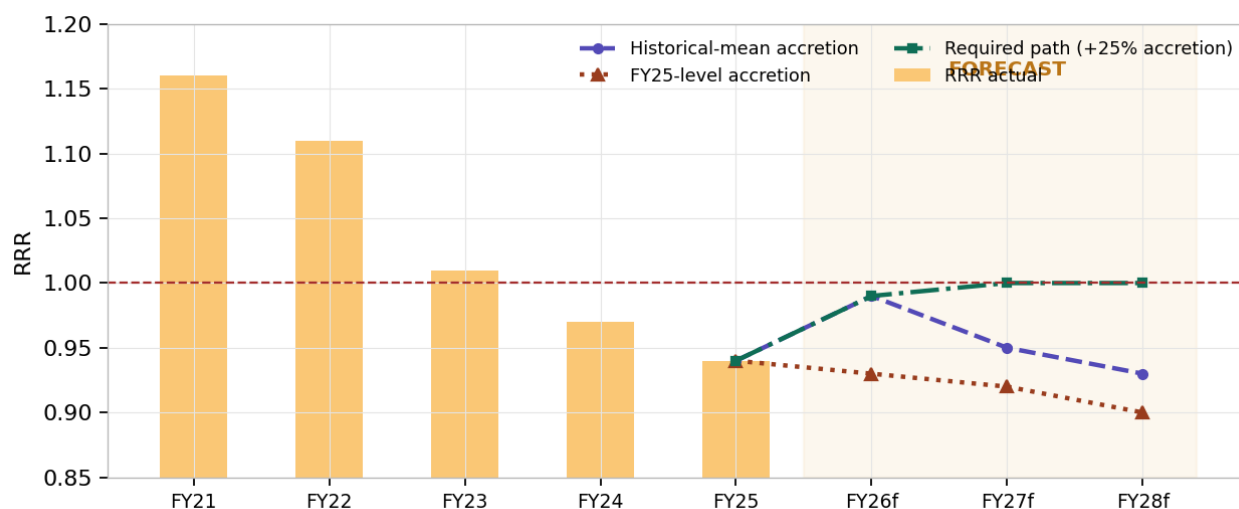


Model: $\text{Production}(t) = \beta_0 + \beta_1 \cdot \text{Workovers}(t) + \beta_2 \cdot \text{Workovers}(t-1) + \beta_3 \cdot \text{DevWells}(t-1) + \varepsilon$, with β_0 replaced by an Arps decline baseline in the full version. The shaded gap between the scenario lines is the production that intervention buys. Naive eyeball estimate from the data: ~4,100 tonnes per incremental workover (FY21–25: +121 workovers, +0.50 MMT).

SME checks: ① Plausible decline rate for mature Assam fields (5–8%/yr assumed)? ② Are FY27 scenario anchors (3.48 sustained / 3.33 reduced) directionally sane? ③ Confounders to control: gas-lift optimisation, EOR pilots, post-COVID normalisation.

4 RRR scenario fan — FY26–28

Predictive claim #2 · Monte Carlo accretion scenarios (3 deterministic paths shown) · ILLUSTRATIVE

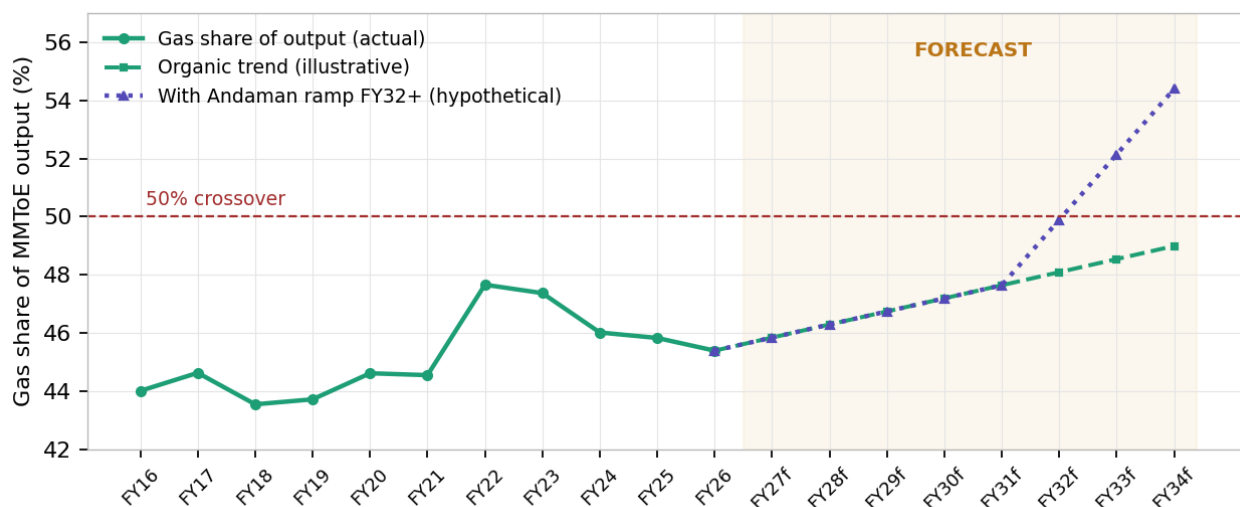


Claim: accretion must rise ~25% above the historical mean to hold RRR at 1.0 by FY28. Note: FY26 accretion (6.23 MMT0E) is already an actual in the 10-yr Excel — FY26 RRR may be computable rather than forecast, pending confirmation of the matching depletion figure.

SME checks: ① Which depletion basis pairs with the accretion series? ② Is the +25% requirement consistent with the sanctioned exploration programme? ③ Confirm FY26 RRR from the FY26 annual report when published.

5 Gasification crossover projection

Predictive claim #4 · gas share of MMToE output crossing 50% · ILLUSTRATIVE, conversion 1 BCM ≈ 0.9 MMToE

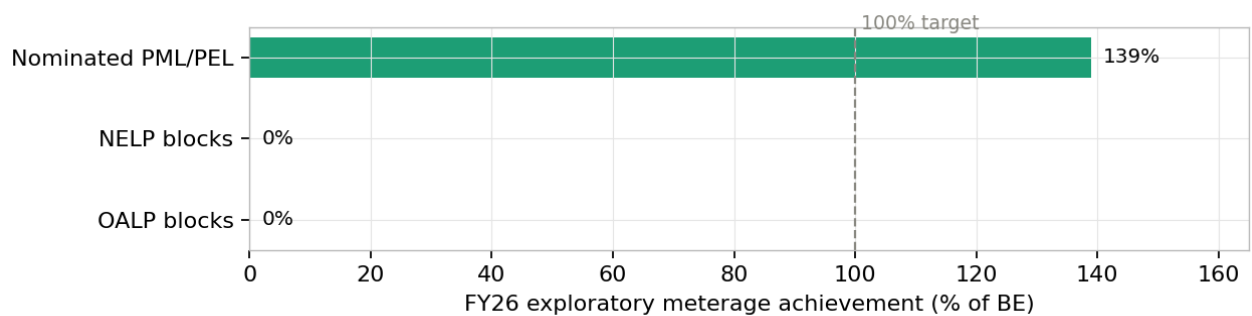
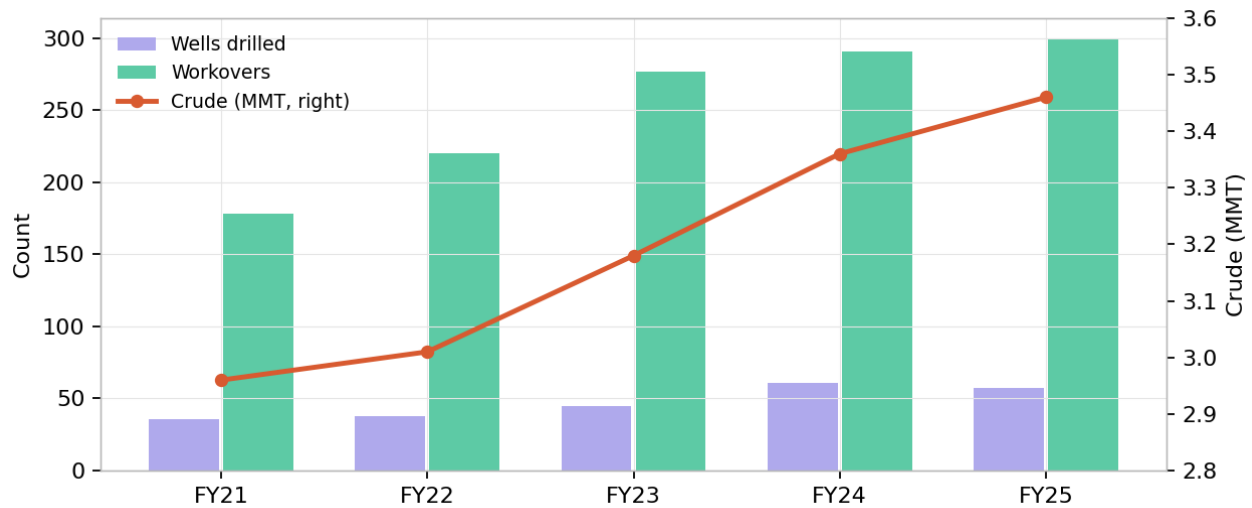


Claim: gas is already ~45% of output on an energy basis. Organic drift alone approaches 50% around FY32–34; a materialised Andaman ramp (hypothetical, FY32+) pulls the crossover earlier and steepens it. Next layer: join quarterly realisation data to convert the volume shift into a revenue-mix forecast.

SME checks: ① Preferred gas-to-MMTToE conversion factor? ② Organic drift rate (+0.45 pp/yr assumed) sane vs sanctioned projects? ③ Should LPG/condensate enter the output mix?

6 Exploration page — KPIs & analytics

Wells drilled (FY25) 57 vs 36 in FY21 (+58%)	Expl. meterage ach. (FY26) 139% nominated NELP / OALP blocks: 0%	Workover operations 299 FY25 actual · FY26 plan 307	Reserve accretion (FY26) 6.23 MMT_oE +6.1% · best in 5 years
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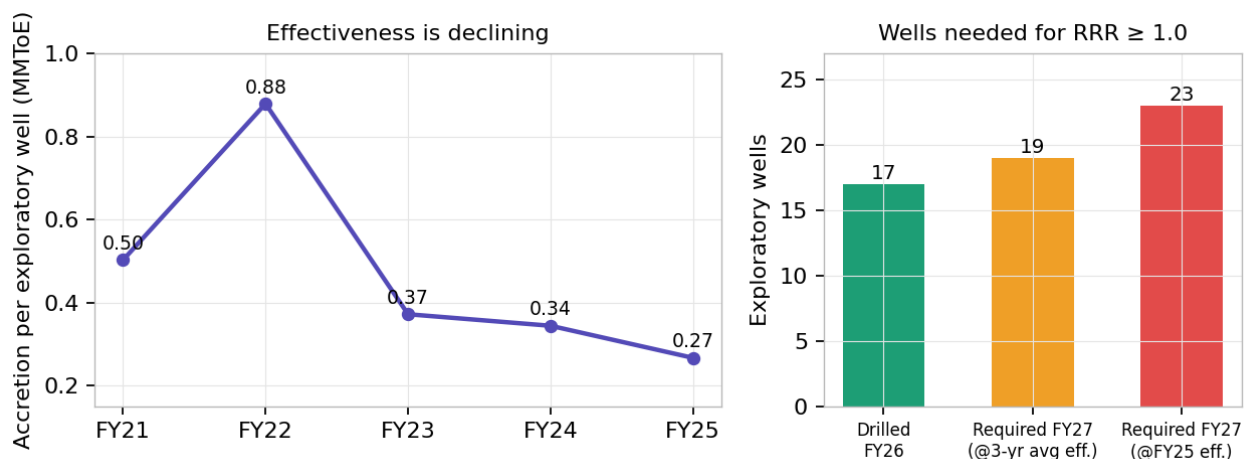


Insights: intervention intensity up 58% (wells) / 68% (workovers) in five years, coincident with the production recovery — the regression in section 3 prices this. Execution is concentrated in legacy nominated acreage; NELP/OALP onshore blocks at 0% meterage are the commitment-risk watch-item, while the single offshore campaign (Andaman) delivered 2-of-3 gas-bearing wells.

SME checks: ① FY24 wells (61) exceed FY25 (57) — does the causal framing survive the pullback? ② Were NELP/OALP wells deferred for legitimate reasons (rig scheduling, clearances) rather than execution failure? ③ Confirm FY25 split: 21 exploratory + 35 development + 1 offshore.

7 Exploration effectiveness & required-wells inversion

Predictive claim #7 · accretion per exploratory well, inverted to an FY27 requirement · ILLUSTRATIVE



Claim: accretion per exploratory well has fallen from ~0.50 to ~0.27 MMTToE (FY21–25) — more wells are buying less reserve. Inverting at current effectiveness: FY27 needs ~19–23 exploratory wells to deliver $RRR \geq 1.0$, vs 17 drilled in FY26. This panel cross-references the RRR fan (section 4): the two pages solve the same equation from opposite ends.

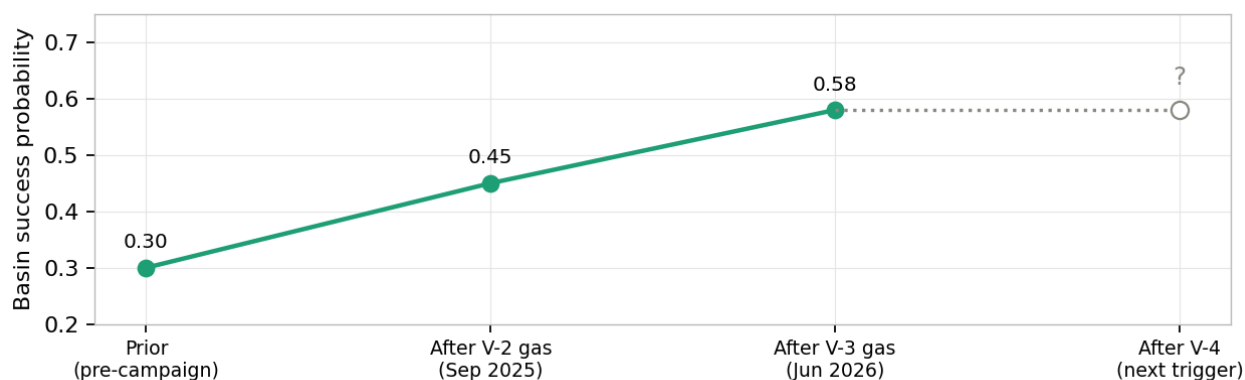
SME checks: ① Is per-well effectiveness a fair metric, or should it be per-metre / per-Rs-capex? ② Accretion includes revisions and EOR additions, not just exploration — should the numerator be split? ③ Is the declining-effectiveness story consistent with field maturity expectations?

8 Andaman scenario module — possible upside

SCENARIO · ILLUSTRATIVE · UNBOOKED — excluded from all 2P, accretion and RRR computations. No official reserve estimate exists; volumes below are analog illustrations, not estimates.

Vijayapuram-2 · Sep 2025 Gas · 87% methane · 295 m WD	Vijayapuram-3 · Jun 2026 Gas · continuous flaring · 355 m WD	Next Appraisal → FID	First gas ~FY32+ frontier norm 7–10 yrs
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Bayesian basin-probability tracker (predictive claim #9) — prior updated on each well result; values below are illustrative placeholders for SME calibration:

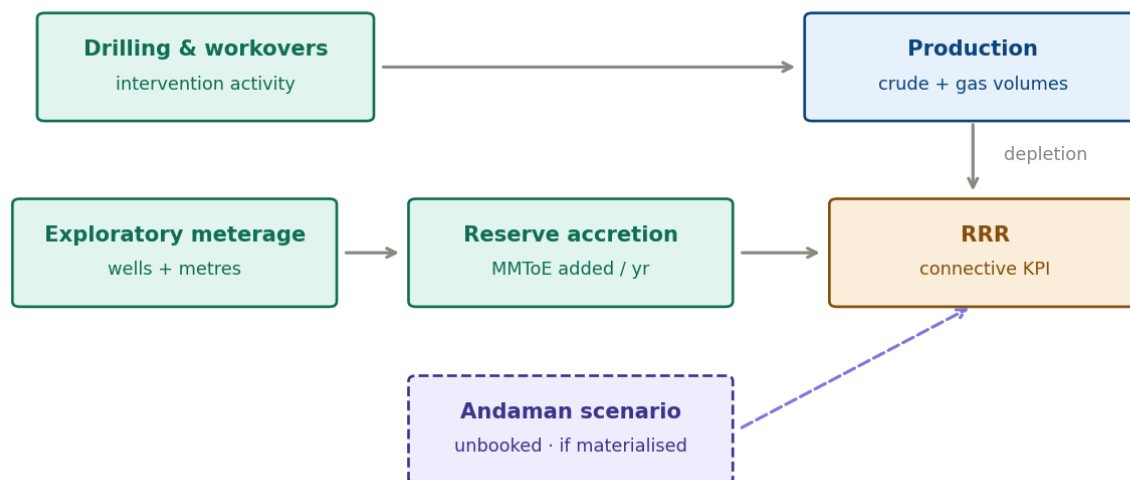


If materialised — hypothetical booking vs current 2P gas of 139 BCM (claim #10):

	Small analog · 25 BCM	Mid analog · 50 BCM	Large analog · 75 BCM
Uplift to 2P gas	+18%	+36%	+54%
Reserve life added (at current prod.)	≈ +8 yrs	≈ +15 yrs	≈ +23 yrs

SME checks: ① Replace 25/50/75 BCM with geologically comparable analog fields — or strike the panel if any number is premature. ② Calibrate the prior (0.30 shown) and update sizes. ③ Confirm V-1 status wording (no commercial show announced).

9 Cross-page predictive logic map



RRR is the connective KPI: it appears on the Production page as an outcome, on the Exploration page as the target the accretion model solves for, and in the Andaman module as the number the scenario could reset. Solid arrows are fitted/derivable relationships; the dashed arrow is scenario-only and never enters KPI math (booked = false enforcement in the data schema).

SME check: is this causal map complete — should gas realisation/pricing, capex, or rig availability be first-class nodes?

10 Predictive claims inventory & review checklist

#	Predictive claim	Tier	Shown in
1	FY27 production forecast (decline + intervention regression)	Fitted model (pending)	\$3
2	Monte Carlo RRR trajectory — accretion +25% needed by FY28	Scenario simulation	\$4
3	Reserve Life Index: oil ~20 yrs, gas ~43 yrs	Derived arithmetic	\$2
4	Gas share of output crossing 50% (organic vs Andaman)	Scenario simulation	\$5
5	Intervention ROI: tonnes per workover / per dev well, inverted to FY27 plan	Fitted model (pending)	\$3, \$6
6	NELP/OALP commitment-risk scoring	Scenario simulation	\$6
7	Exploration effectiveness → wells needed for RRR ≥ 1.0	Fitted model (pending)	\$7
8	Bayesian basin probability, updated per Andaman well	Scenario simulation	\$8
9	Hypothetical booking sensitivities (25/50/75 BCM analogs)	Scenario simulation	\$8
10	First gas ~FY32+ (frontier timeline norm)	Derived arithmetic	\$8

Credibility strategy

Lead with fitted models validated against the known FY26 year before any external showing; present scenario simulations with assumptions visible and bands shown; never label derived arithmetic as machine learning. The Andaman module's "unbooked / illustrative" badging is the template for all scenario content.

Consolidated SME review checklist

- Verify RRR definition (annual vs 3-yr rolling) and the depletion basis pairing with accretion.
- Confirm FY26 figures: crude 3.45 MMT / gas 3,186 MMSCM / accretion 6.23 MMTtoE / achievement 90.9% & 87.1%.
- Pressure-test the workover→production causal story against gas-lift, EOR and normalisation confounders.
- Rule on FY24 (61 wells) vs FY25 (57): does the intensity narrative hold?
- Classify the NELP/OALP 0% as deferral vs execution risk.
- Replace Andaman analog volumes (25/50/75 BCM) with SME-chosen analogs or strike the panel.
- Calibrate the Bayesian prior and update steps; confirm Vijayapuram-1 status wording.
- Approve conversion factors: 1 BCM $\approx$ 0.9 MMTtoE; gas share methodology; LPG/condensate inclusion.
- Confirm decline-rate prior (5–8%/yr) for mature Assam fields.
- Decide effectiveness denominator: per well, per metre, or per Rs capex.